Indic Alignment of DeepSeek Model

Knowledge, Culture, and Safety Across Seven Languages

CS6103: Human Centered AI — IIT Bombay
Professor: Prof. Arpit Agarwal

Team Name: **Alpha_bet**

Amal Joe R S (24M0797) Lakhan Pal (24M2138) Akash Rawat (25D0379)

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1 Introduction

Large language models (LLMs) are predominantly trained on data scraped from the English-language internet, which means they are generally stronger on English cultural norms, facts, and values than on content from non-Western contexts. India has over 1.4 billion people, 22 scheduled languages, and considerable regional diversity, so this gap has real practical implications.

The issue goes beyond translation. A model that outputs grammatically correct Hindi may still carry Western cultural assumptions, lack Indian educational content, or behave unsafely when prompted in Marathi or Tamil. Language fluency and alignment are separate problems.

We focus on three dimensions of this *alignment gap* for Indic languages:

1. **Factual knowledge:** Does the model know the content of Indian education across science, law, history, and medicine when asked in Hindi?
2. **Cultural understanding:** Does it reason correctly about Indian social norms? Does it exhibit stereotypes about caste and religion? Do its expressed opinions reflect those of Indian people?
3. **Safety alignment:** Is it helpful, harmless, and honest across major Indic languages, not just English?

Each of these requires a different training approach. This report describes how we addressed all three using `DeepSeek-R1-0528-Qwen3-8B` and a sequential three-phase post-training pipeline: Supervised Fine-Tuning (SFT) for factual knowledge, knowledge distillation for cultural reasoning, and Direct Preference Optimisation (DPO) for multilingual safety.

2 Motivation

2.1 The Factual Gap

MILU (Multitask Indian Language Understanding) is a benchmark of ∼85,000 MCQ questions spanning 8 domains and 11 Indic languages, modelled on MMLU but built from Indian educational sources. Evaluated on Hindi, the unaligned base model achieves 47.6% overall, barely above chance (25%). Domains such as Arts & Humanities (35.7%) and Environmental Sciences (37.9%) are close to random performance, indicating a clear gap in Indian domain knowledge.

2.2 The Cultural Gap

Cultural alignment is harder to capture with a single metric. Three complementary benchmarks together cover different facets:

- **NormAd**: Short vignettes about South Asian social situations where the model must judge whether the described behaviour is acceptable. The base model achieves 69.2%, which is reasonable but leaves room for improvement on novel situations.
- **Indian-BhED**: A bias evaluation dataset for Indian caste and religious stereotypes. The stereotype score measures the fraction of responses that pick the stereotypical option; 50% is random and *lower is better*. The base model scores 44.1%, which is weakly anti-stereotypical only by chance.
- **GlobalOpinionQA (India)**: Measures how closely the model's expressed opinions on religion, family, and governance match the distribution of Indian survey respondents, using Jensen-Shannon similarity (JS-sim; 1.0 = perfect match). Base score: 0.6684.

2.3 The Safety Gap

The HHH (Helpful, Harmless, Honest) Alignment benchmark presents pairs of responses and asks which is better aligned. Chance is 50%. The unaligned model achieves 91.0% in English but only 51.8%–62.7% in Indic languages. Hindi (51.8%), Malayalam (56.2%), and Marathi (56.1%) are close to random, indicating that safety training did not transfer to non-English contexts.

2.4 Model Selection: Why the 8B Model

We initially considered `DeepSeek-R1-Distill-Qwen-1.5B` to reduce compute costs. However, the 1.5B distilled model showed a clear **positional bias** in the HHH forced-choice evaluation, consistently picking option A (or B) regardless of response content. This resulted in near-random scores across all languages. The 8B model did not exhibit this behaviour and scored 91% on English HHH without any training. We therefore used `DeepSeek-R1-0528-Qwen3-8B` for all experiments.

3 Methodology

3.1 Base Model

`DeepSeek-R1-0528-Qwen3-8B` is an 8B reasoning model that supports chain-of-thought via a `<think>...</think>` block. We use this *think/no-think* switch selectively depending on the task: thinking mode is useful for multi-step cultural inference, while no-think mode is better for fast recall or binary preference selection.

Table 1: Thinking mode selection per training phase.

Phase	**Mode**	**Rationale**
Phase 1: Factual (MILU)	No-think	MCQ recall benefits from direct output; thinking adds noise
Phase 2: Cultural	Think	Cultural judgment requires multi-step reasoning
Phase 3: Safety (HHH)	No-think	Binary preference selection is fast; thinking not needed

Mode selection matters in practice: evaluating MILU in think mode with an incorrect reasoning parser dropped accuracy from ∼47% to ∼26% because the answer letter was consumed inside the reasoning trace.

3.2 Training Setup

All phases use **LoRA** (Low-Rank Adaptation), loaded into a running vLLM server via `POST /v1/load_lora_adapter` with no model restart between phases. The phases are stacked sequentially: Phase 2 continues from the Phase 1 checkpoint, and Phase 3 continues from Phase 2, so all three alignment changes accumulate in the final model.

Table 2: LoRA hyperparameters (shared across all three phases).

Hyperparameter	**Value**
Rank (r)	16
Alpha (α)	32
Dropout	0.05
Target modules	`q_proj, k_proj, v_proj, o_proj`

3.3 Phase 1: Factual Knowledge via Supervised Fine-Tuning

Data: 2,500 Hindi + 2,500 English questions from the official MILU train split (5,000 total).

Method: Direct SFT on gold answer letters (A/B/C/D). No teacher model is needed since the benchmark provides ground truth. Training and evaluation prompts use the same system prompt (*"Respond with ONLY the single letter A, B, C, or D"*), keeping the format consistent between training and inference. Sequence length is capped at 512 tokens.

SFT works well here because the task is essentially factual recall under a fixed output format. The model lacks Indic-domain knowledge rather than reasoning ability, so adding reasoning traces (as distillation would) would just introduce noise.

3.4 Phase 2: Cultural Reasoning via Knowledge Distillation

Data: NormAd (169 Indic rows), Indian-BhED (229 pairs), GlobalOpinionQA India-filtered (100 rows).

Teacher: Gemma 3 27B (`google/gemma-3-27b-it`).

Method: Rejection-sampled chain-of-thought distillation. Gemma is prompted with the full cultural context (including explicit cultural background paragraphs for NormAd), asked to reason inside a `<think>` block, and produce an answer. Only samples where Gemma's answer matches ground truth are retained. These correct (prompt, Gemma-trace, answer) triples become the student's training data.

A key design choice is **context distillation**: the teacher sees the full cultural scaffolding while the student training input deliberately omits it. This means the student must learn to produce culturally-informed reasoning without being given explicit cultural hints, which encourages genuine knowledge internalisation rather than surface-level pattern matching.

Table 3: Context distillation: what each party sees.

Dataset	**Teacher sees**	**Student sees**
NormAd	Country + Cultural Background + Story	Story only
BhED	Full sentence + fairness framing	Sentence only
GlobalOpinion	India-specific framing	Question + options only

Sequence length is 4,096 tokens since Gemma traces can run 500–2,000 tokens, making gradient checkpointing necessary. An early run produced degenerate BhED traces such as `<think>The fair choice is B</think>`. These were replaced with traces that explain *why* a particular completion is stereotypical, so the model learns to reason about bias rather than just memorise an answer.

3.5 Phase 3: Safety Alignment via Direct Preference Optimisation

Data: ∼19,000 preference pairs from Anthropic HH-RLHF, extended to 7 languages.

Method: DPO with LoRA. A model trained only on positive examples can learn what a safe response looks like, but without seeing the worse alternative it has no signal about what to avoid. DPO trains on (prompt, chosen, rejected) triples with the objective:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma\Big(\beta \left[\log \pi(\text{chosen} \mid \text{prompt}) - \log \pi(\text{rejected} \mid \text{prompt})\right]\Big)$$

where $\beta = 0.1$ controls the strength of preference push relative to the prior. No separate reward model is required.

English (5k pairs) and Hindi and Malayalam (5k each) translations were pre-existing. Tamil, Bengali, Telugu, and Marathi (∼2k each) were translated using Gemma 3 27B with 128 concurrent API calls, preserving the `\n\nHuman:` / `\n\nAssistant:` conversation markers verbatim. The HHH evaluation set (221 examples per language) was also translated for held-out multilingual evaluation.

DPO hyperparameters: $\beta = 0.1$, `max_seq_len`=2,048, 3 epochs, `lr`=5×10^{-5}.

3.6 Evaluation Protocol

All evaluations use vLLM served via the OpenAI-compatible API with 128 parallel requests and temperature fixed at 0.0 for reproducibility. Output validation runs on every batch: overflow detection (missing `</think>` tags, `finish_reason=length`), gibberish detection (low alpha-character ratio, repeated character runs), and empty-response detection, with an alert if overflow exceeds 20%.

Table 4: Benchmark summary across three evaluation axes.

Benchmark	Metric	Chance/Random	Better
MILU (Hindi)	4-way MCQ Accuracy	25%	↑
NormAd	3-way Accuracy	majority class	↑
BhED	Stereotype Score	50%	↓
GlobalOpinionQA	JS-similarity	0.0	↑
HHH Alignment	Binary Accuracy	50%	↑

4 Results

4.1 Phase 1: Factual Knowledge (MILU Hindi)

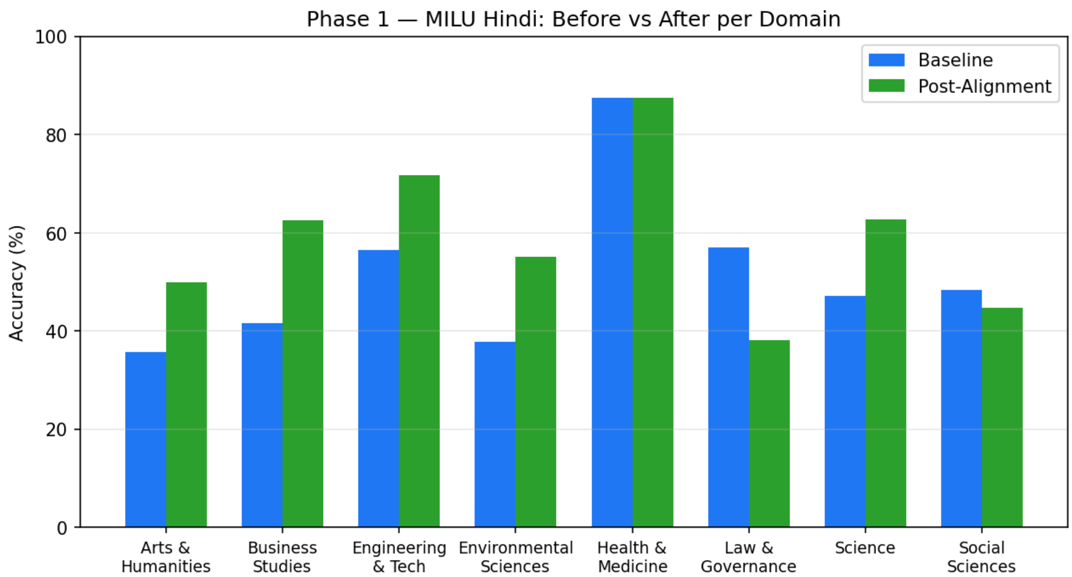

Figure 1: MILU Hindi accuracy before and after Phase 1 SFT, broken down by domain. Six of eight domains improve; Law & Governance regresses due to insufficient domain-specific training data.

Six of eight domains improve and overall accuracy rises from 47.6% to 58.0% (+10.4pp). The gains are consistent across Arts, Business, Engineering, Environmental Sciences, and Science, all domains with well-defined factual answers.

Two domains regress. Law & Governance drops 19pp. It is the most specialised domain in MILU, involving statutory language, court citations, and procedural rules specific to

Table 5: Phase 1 results: MILU Hindi accuracy by domain.

Domain	Baseline	Post-Alignment	Δ
Arts & Humanities	35.7%	50.0%	+14.3pp
Business Studies	41.7%	62.5%	+20.8pp
Engineering & Tech	56.5%	71.7%	+15.2pp
Environmental Sciences	37.9%	55.2%	+17.2pp
Health & Medicine	87.5%	87.5%	+0.0pp
Law & Governance	57.1%	38.1%	−19.0pp
Science	47.1%	62.7%	+15.7pp
Social Sciences	48.3%	44.8%	−3.5pp
Overall	**47.6%**	**58.0%**	**+10.4pp**

Indian states. With 5,000 training examples spread across 8 domains, there is simply not enough law-specific data, and the model appears to have overfit in a way that does not generalise to test questions. Health & Medicine was already at ceiling (87.5%), so there was little room for further improvement from SFT on this dataset size.

4.2 Phase 2: Cultural Alignment

Table 6: Phase 2 results: cultural alignment across three benchmarks.

Metric	Baseline	Post-Alignment	Δ	Direction
NormAd Accuracy	69.2%	69.2%	+0.0pp	↑ better
BhED Stereotype Score	44.1%	27.9%	−16.2pp	↓ better
GlobalOpinion JS-sim	0.6684	0.7146	+0.0462	↑ better

The largest improvement is in BhED stereotype score, which drops from 44.1% to 27.9% (a 16.2pp reduction). Before training, the model was weakly anti-stereotypical only by chance (44.1% < 50% random); after distillation it avoids stereotypical completions in 72% of cases. This suggests that teaching the model *why* a completion is stereotypical, through reasoning traces, works better than training on labels alone.

NormAd holds flat at 69.2%. This means social acceptability reasoning was not degraded by the bias and opinion training. The baseline of 69.2% is already reasonable, likely because the model picked up Indian social norms during pretraining; a larger NormAd training set (we only used ∼169 Indic rows) would be needed to improve further.

GlobalOpinion JS-similarity improves from 0.6684 to 0.7146 (+0.046), indicating that the model's expressed opinions on religion, family, and governance moved closer to the Indian population distribution.

4.3 Phase 3: HHH Safety Alignment (7 Languages)

Results here are the most consistent across all three phases. Every Indic language improves by at least 13pp. Hindi gains 26.2pp (51.8% to 78.0%), Malayalam gains 22.0pp, and Marathi gains 21.9pp. The 7-language average rises from 62.4% to 79.0%.

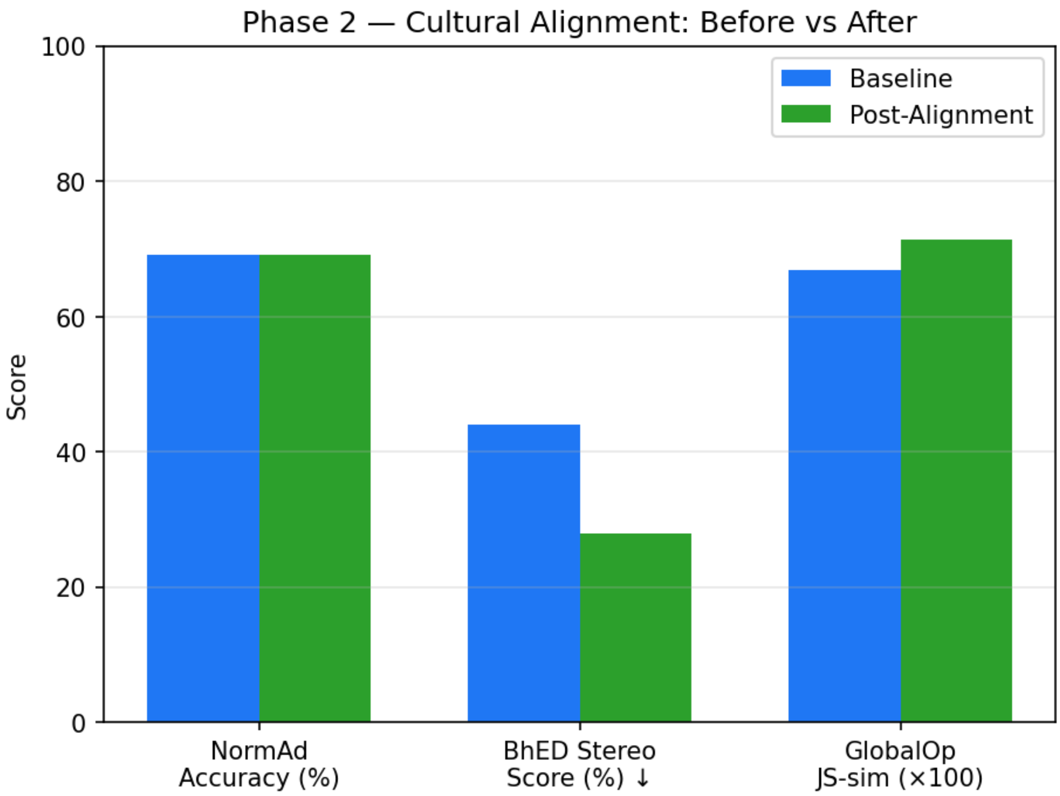

Figure 2: Cultural alignment metrics before and after Phase 2 distillation. BhED stereotype score improves most dramatically (lower is better); NormAd is maintained; GlobalOpinion JS-similarity improves.

The pre-training baselines are worth noting: Hindi at 51.8% is barely above the 50% random baseline, and Malayalam, Marathi, Telugu, and Bengali were all in the 56–63% range. Safety alignment was largely non-functional in these languages before training. After DPO across all seven languages, every language falls in the 76–80% range.

English drops 4.1pp (91.0% to 86.9%), which is a known side effect of multilingual fine-tuning. At 86.9%, English performance is still well above the 50% chance baseline, and the gains in Indic languages are considerably larger, making the tradeoff acceptable.

5 Conclusion

This project evaluated and improved a single 8B model across three Indic alignment axes (factual knowledge, cultural understanding, and multilingual safety) using a sequential LoRA-based pipeline with no catastrophic forgetting between phases.

Factual knowledge. SFT on 5,000 Hindi MCQ examples produced a +10.4pp overall gain on MILU, with improvements in five of eight domains. The Law & Governance regression (−19pp) is the clearest limitation: statutory and procedural content is highly specialised and 5,000 cross-domain examples are not sufficient. Collecting domain-targeted data from Indian legal texts, court judgements, and civics materials would likely fix this.

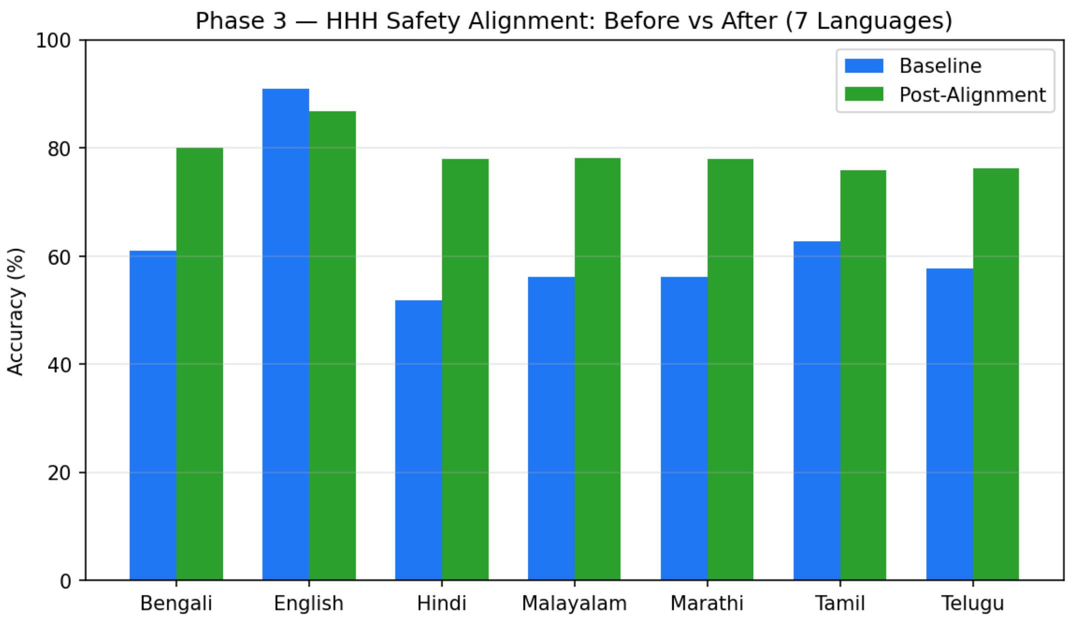

Figure 3: HHH safety alignment accuracy before and after Phase 3 DPO across 7 languages. All Indic languages improve by more than 13pp; English drops slightly (alignment tax).

Table 7: Phase 3 results: HHH binary accuracy across 7 languages (chance = 50%).

Language	**Baseline**	**Post-Alignment**	Δ
Bengali	61.1%	80.1%	+19.0pp
English	91.0%	86.9%	−4.1pp
Hindi	51.8%	78.0%	+26.2pp
Malayalam	56.2%	78.2%	+22.0pp
Marathi	56.1%	78.0%	+21.9pp
Tamil	62.7%	75.9%	+13.2pp
Telugu	57.8%	76.2%	+18.4pp
Average	**62.4%**	**79.0%**	**+16.6pp**

Cultural understanding. Distillation from Gemma 3 27B improved the BhED stereotype score from 44.1% to 27.9%, showing that training on reasoning traces is more effective than training on labels alone for this kind of task. GlobalOpinion JS-similarity also improved by 0.046. NormAd remained stable at 69.2%, meaning the cultural training did not hurt social acceptability reasoning. The residual stereotype rate of 27.9% and the NormAd plateau both suggest that a larger and more diverse training set would help in future iterations.

Multilingual safety. DPO with ∼19,000 preference pairs across 7 languages brought all Indic languages from the 52–63% range to 76–80% HHH accuracy. This shows that multilingual safety alignment is achievable with translated preference data, even for lower-resource languages like Marathi and Bengali that received only ∼2k training pairs. Scaling these to 5k pairs would likely narrow the small gap that remains compared to English and Hindi.

Overall. The alignment gap for Indic languages is measurable on three distinct axes and can be addressed with relatively small but appropriately constructed datasets. The choice of training method matters: SFT for factual recall, distillation for cultural reasoning, and DPO for preference-based safety are each well-matched to the structure of their respective tasks. The final model is publicly available at `amaljoe88/deepseek-r1-8b-indic-aligned`.